Non-Destructive Testing

Non-destructive testing methods are testing methods that do not cause destruction nor impair its intended use. There are 7 common NDT methods:

Visual Inspection

It is the most common, cheapest and versatile method, for surface inspections. Usually, the area is viewed with the unaided eye, but other instruments like magnifying glasses, borescope, portable video inspection units and robotic crawlers.

Hydrostatic Testing

Hydrostatic testing is used for checking for leakage of components, more specifically components that are some form of vessel. The defects found are through holes and not pores, and are found by the flow of gas or liquid through the leak locations, for example with water and air bubbles, or helium and helium leak detectors.

Generally, the procedure involves filling the vessel with a respective fluid and pressurising the fluid to 1.5 to 2 times the working pressure observe any leakage, indicating the discontinuity locations. For a more sensitive test, gas can be applied but the setup must be immersed in water to observe the gas leaks as bubbles in the water. The most sensitive type of hydrostatic testing is using helium, due to its small and inert gas particles.

Although the method works with all materials, the high pressure during the procedure may cause cracks in porous materials, thus it is more reliable in non-porous materials.

Dye Penetrant Testing

This method is used for checking for small cracks or discontinuities too small for visual inspection, however is only applicable to surface defects open on a homogenous material that is non-porous and non-laminate. It is also cheap and does not require expensive equipment.

The two main types of penetrants are visible dye; a brightly coloured dye like red that is viewed under normal white lighting, or fluorescent dye which can glow under UV light.

They also need to have their excess removed, and can be categorised by their removal method; water washable, which is the least sensitive type of

penetrant, post-emulsified, where an emulsifier is required to emulsify the oil based dye to be washed off with water, and solvent-removable, where a specific solvent is required to dissolve the excess.

The dye penetrant testing procedure is as follows:

Surface is cleaned

Penetrant is applied

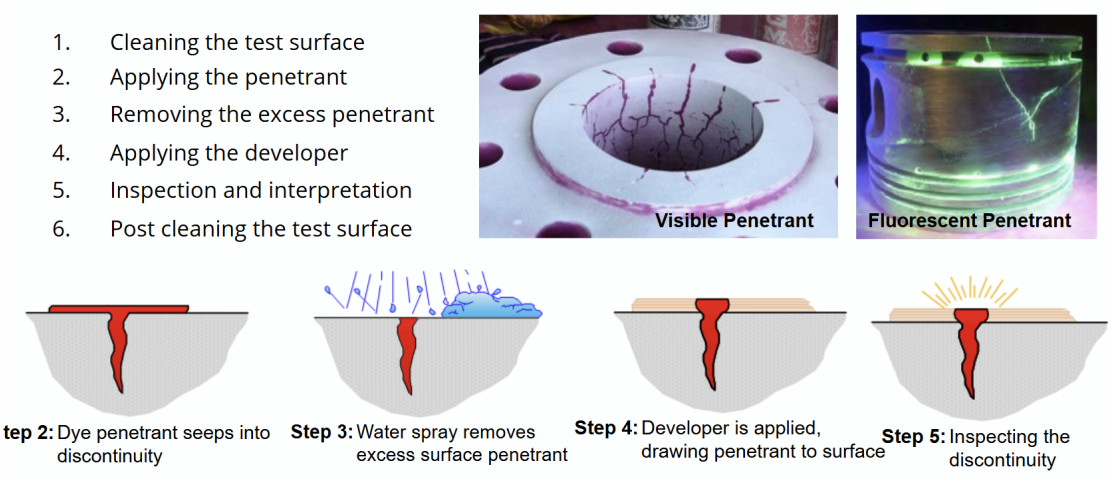
Excess penetrant has emulsifier or solvent applied, if applicable, to dissolve the excess

Dissolved excess is washed off with water

A developer in either wet or dry form is applied, either via a powder-water solution or as a dry powder respectively, which draws the penetrant to the surface for good contrast view.

Inspect can then be conducted, with the respective light needed

Post cleaning is done afterwards to remove any chemicals that may interfere with other processing or service use.

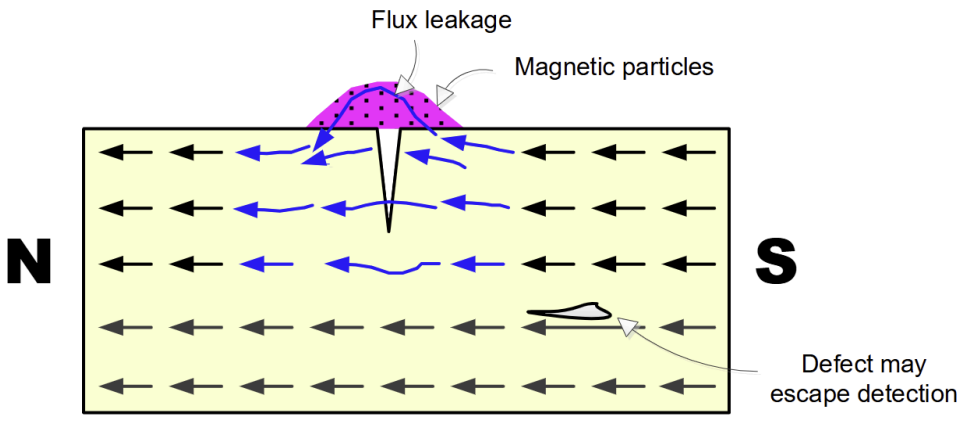


Magnetic Particle Testing/Magnetic Particle Inspection

This method is used for surface or sub-surface defects in ferromagnetic materials.

Recall that when a ferromagnetic item is placed in a magnetic field, magnetic flux flows along the magnetic field through the ferromagnetic item. In an item with no defects, the magnetic flux is uniform. However, with defects, the magnetic flux is no longer uniform and there is a flux

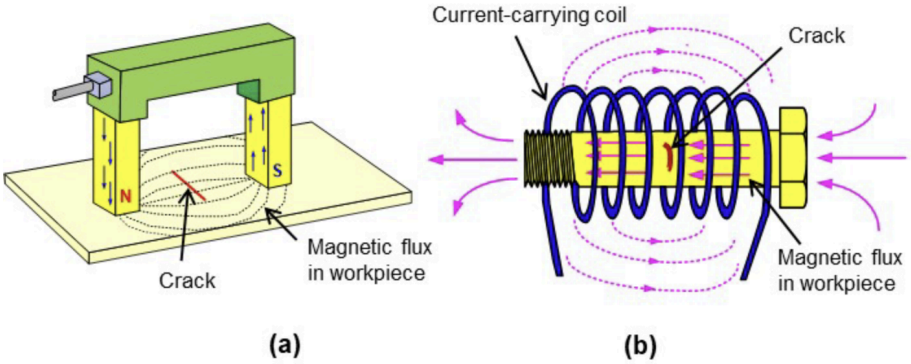
leakage at the crack. When finely divided magnetic particles are then applied to the magnetised item, they are attracted and concentrate around the flux leakage, showing where the cracks and defects are.



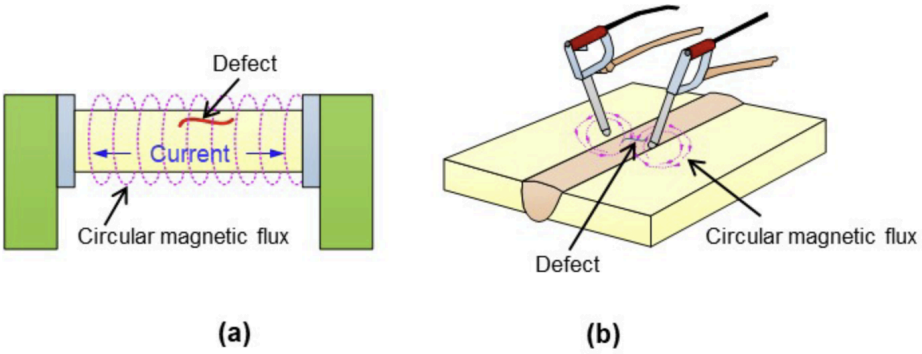
However, if the defect is aligned in the same direction as the flux, flux leakage may be minimal and the crack will escape detection. Thus, cracks must lie between 90 to 45 deg to the magnetic field lines to be detected.

The procedure is as follows:

- Material undergoes pre-cleaning
- Magnetic fields are introduced
  - There are different methods to apply magnetic fields, and in different directions.
  - Generally, there are two ways to apply magnetic fields; longitudinally (along a straight direction, e.g. along the poles for circular parts) and circumferentially (along a circular path)
  - For longitudinal magnetic fields, they can be formed with permanent magnets/yoke electromagnets by placing the poles on the workpiece to have the flux flow from one pole to the other, ideally normal to the crack direction, or with a solenoid, to have current run in a coil around the part to induce a horizontal flux along the solenoid, thus having magnetic flux pass through the component. These methods are best applied for transverse cracks.



Circumferential magnetic fields can be formed by direct induction, where current is passed directly through the component via electrical end contacts to induce a circular coil of magnetic fields, or by prod contacts, where larger components that cannot have end contacts applied OR components that are not rods have prod contacts applied instead to induce a circular magnetic field. These methods are best applied for longitudinal cracks.



Thus, magnetic fields are typically applied in two (usually perpendicular) directions in each pass to ensure any cracks have magnetic flux flow in a non-parallel direction to them.

Apply magnetic particles

Magnetic particles can be applied in either a dry or wet form.

Dry magnetic particles, which may be brightly coloured for better contrast, is applied with a hand shaker, spray bulb, or air stream, forming a light dusting of the particles.

Wet magnetic particles are applied in an oil suspension, and again are coloured/fluorescent for good contrast. Generally, these

particles cause a more sensitive test as the oil suspension allows particle mobility. Usually, the wet particles are applied via dipping, immersing, or spraying

Inspection & interpretation

Demagnetisation

Even when the magnetic flux is no longer applied, residual magnetic fields may still be retained, which may interfere with its service. For example, it may interfere with compasses in aerospace applications, chips may be attracted to the part which interferes with machining operations, may attract other magnetic particles which causes increased surface wear, or may induce stray voltages in nearby circuits.

To avoid these effects, demagnetisation is done by applying the magnetic field in the opposite direction, causing the magnetic domains to be misaligned again.

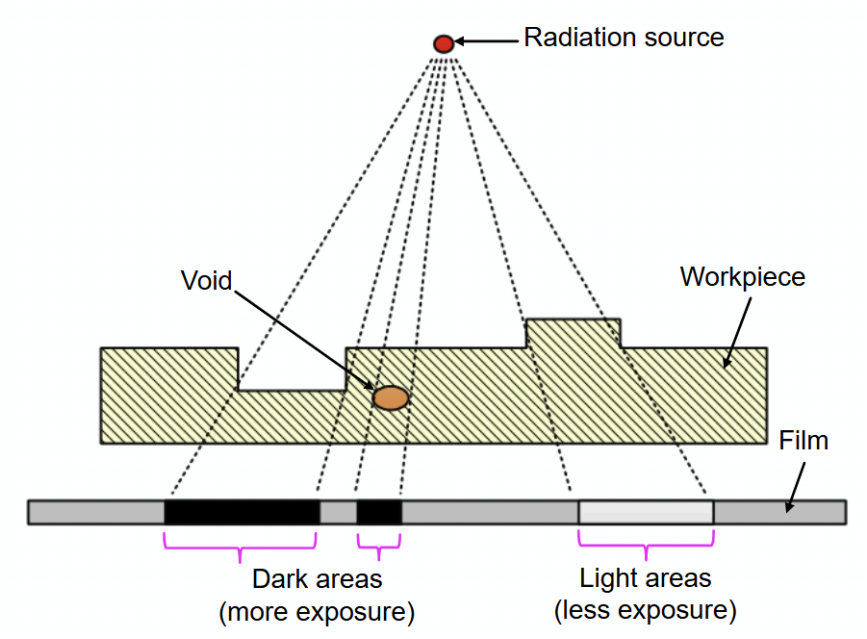
Radiography Testing

This method can be applied for any objects and it images the object's internal structure, for verifying the internal structure and integrity of the specimen. It utilises short-wave EM radiation like X rays and gamma rays and the radiation penetrates the specimen and forms an image on a photographic film or on a fluorescent film. Thus, it is capable of detecting internal defects.

The two main types of radiography is determined by the type of radiation used, X-rays and Gamma rays.

X-rays have a longer wavelength of 0.01 to 10 Angstroms while Gamma rays have a shorter wavelength of 0.005 to 0.1 angstroms, thus causing gamma rays to be more penetrating. However, X rays can be controlled easier as they are produced with an X-ray tube while gamma rays can only be produced by the decay of radioactive isotopes, like radium.

The main working principle of radiography is that the workpiece is placed between a radiation source and a film, and the film's darkness varies based on the workpiece. Thinner sections and voids show up as dark areas on the film, as more radiation can reach the film, while thicker sections show up as lighter areas on the film as less radiation reaches the film.



During radiography, there are different parameters that affect the final image.

Film Density

A measure of the degree of darkness on the film, determined by the amount of radiant energy transmitted through the film, or by the formula

$$D = \log\left(\frac{I_0}{I_t}\right)$$

where I0 is the incident radiation and It is the transmitted radiation.

Film Contrast

A measure of the differences in film density at different sections of the image. Having a higher film contrast means that the image will be more sensitive and less thin defects will still be visible on the film.

Film Definition

Indicates the sharpness of the image, thus higher definition makes it easier to detect defects. The two main factors affecting the film definition are the film unsharpness, dependent on the film quality, and the geometric unsharpness.

Geometric unsharpness is an effect of the setup, thus it can be reduced by:

Increasing the source-specimen distance,

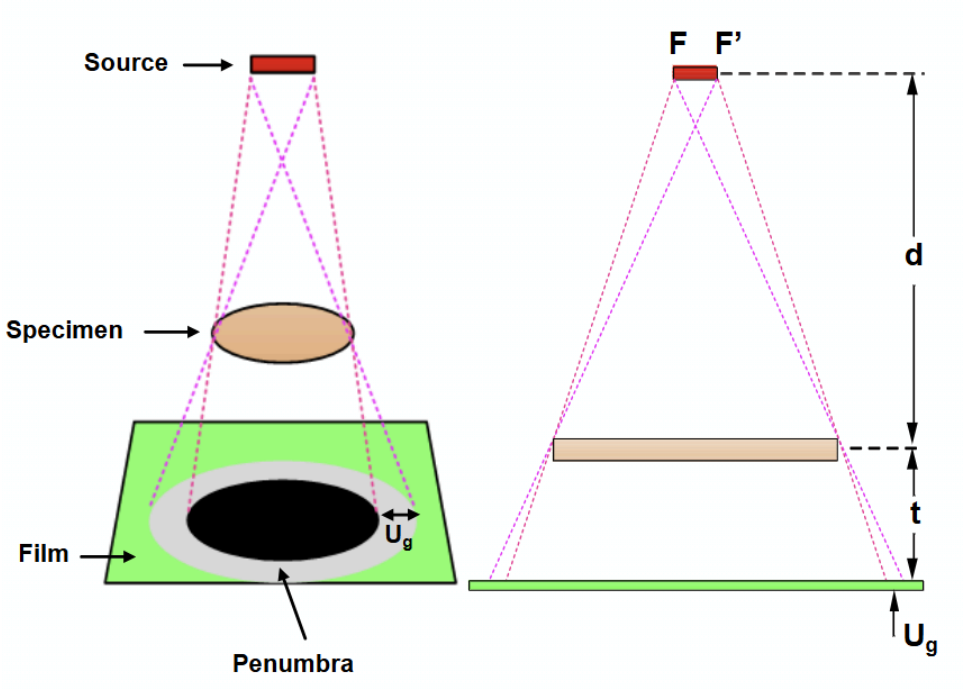
Decreasing the specimen film distance,

Minimising the source width (FF′)

as well as ensuring that

the rays are perpendicular to the film surface

film and specimen lie in parallel planes



Radiographic Sensitivity

A measure of the smallest discontinuity that can be read from the final image.

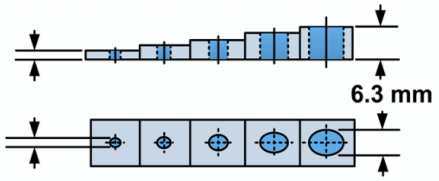
It is a percentage as

$$\% \text{ Sensitivity} = \frac{\text{Thinnest length of defect visible}}{\text{Specimen thickness}} \times 100\%$$

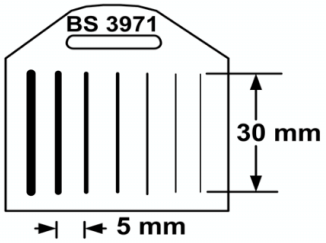
Thus, the smaller sensitivity is, the better it is at detecting small defects.

But usually, we use image quality indicators (IQIs), also known as penetrameters, which come in the form of different standard wire thicknesses, or step holes of different thicknesses and sizes.

- Two types of image quality indicator (IQI):



Step hole IQI



Wire IQI

Ultrasonic Testing

This method can be used for detecting defects that are on the surface, subsurface or internal, by using ultrasound at ~20 kHz, and is applicable all non-laminate materials . Compared to radiography, ultrasound can travel for long distances in the material without losing energy.

Ultrasound is produced with a ultrasonic transducer, containing a piece of piezoelectric material that produces mechanical vibrations, sound, when current is applied, and can also produce current when sound is received. The most common piezoelectric materials are quartz, barium titanate, lithium sulfate, lead zirconate titanate.

Other than the piezoelectric material used, the way the probe interfaces with the surface can also be different. There are two types; normal and angled. A normal probe interfaces perpendicular to the surface, and detects defects parallel to the test surface. Angled probes are placed at an angle to the test surface, for detecting defects not parallel to the test surface. The probes also require a coupling agent between the probe and specimen surface, as any air gaps may cause the ultrasound to be reflected away on the interface, causing poor readings.

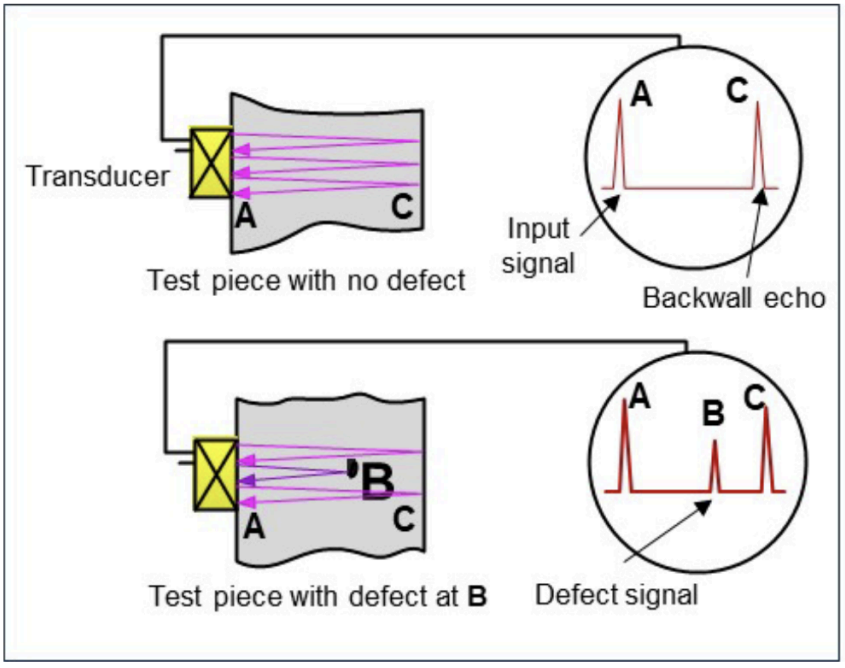
For the ultrasonic testing method itself, there are two types:

Pulse Echo

This method uses only one probe as a transmitter and receiver, and works by sending out a pulse of ultrasound into the material, and waits for it to be reflected against the back surface of the specimen and back to the probe again. With no defects, the received signal after the echo can have the specimen's thickness determined. With defects however, the ultrasound is reflected earlier, and a smaller spike will be observed. The distance on the plot of the ultrasound received can be used to determine the depth of the defect. Thus, this method is able

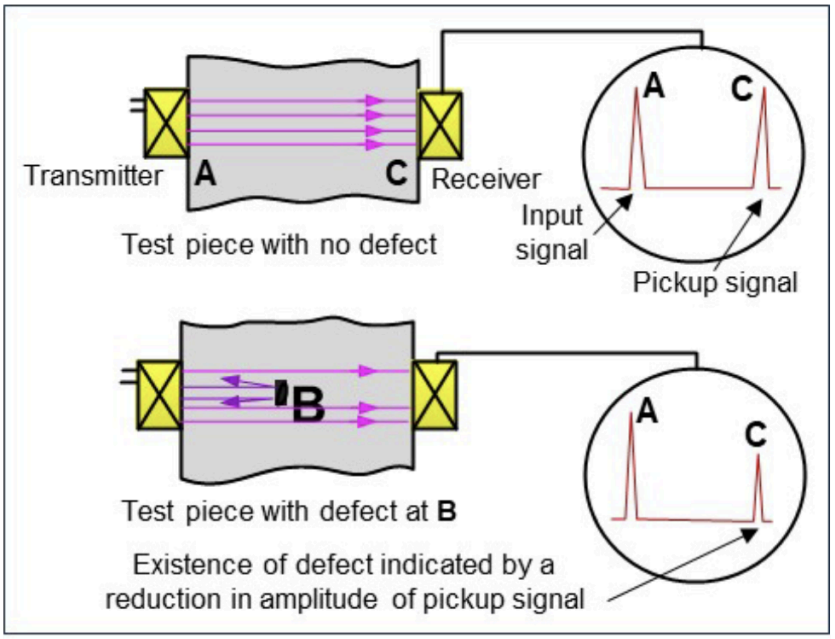
to measure the depth of the defect as well as the wall thickness effectively and accurately.

The pulse echo method also only requires one surface of the sample; useful for when you are unable to access the opposite side of the specimen.



Through Transmission

Through transmission uses 2 probes, one as a transmitter and another as a receiver separately. The main working principle is that the transmitter sends out a pulse of ultrasound for the receiver. As such, the probes need to be on opposite parallel faces and need to be aligned with each other, to ensure the probes can send and receive the ultrasonic pulse.



With no defects, the input signal will be close to the same amplitude as the received signal, but with defects, some ultrasound is reflected away and a lower amount of ultrasound reaches the receiver. Thus, this method only says if a defect is present, but does not give any depth information. This method can measure the specimen thickness, but more work is required to ensure the probes are aligned, thus it is not always accurate.

Eddy Current Testing

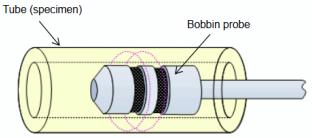
Eddy current testing involves finding surface and subsurface defects on conductive, non-ferromagnetic materials, by using electromagnetic induction to induce eddy currents in the conductive material via AC in a coil. The eddy currents produced causes another magnetic field to be induced in the specimen. With defects, the eddy currents are interrupted and distorts and the magnetic field produced, as well as the output impedance, is reduced. This reduction can be measured and can indicate the defect location, or the metallurgical structure.

Thus, this method can be used for not just defect detection, but also material identification/sorting, measurement of conductive coatings, or measurement of non conductive coatings on a conductive base.

Different types of coils, or probes, can be used depending on the specimen’s shape.

Concentric coils is used for circular extrusions, like rods and bars.

Surface coils is used for flat surfaces, like plates.

Bobbin coils is used for hollow specimens, like pipes and tubes.

Concentric or encircling coil	Surface probe	Inside or bobbin probe
Inspect round bars, rods, wires and tubes	Can be placed on surface of specimen	Can be moved through tubes and pipes